

## In Plain Terms

We're building a map app for people in Canada who live, work, or spend time near forests and grasslands. You tap a spot on the map, like your cabin, your farm, or your town, and the app tells you how a wildfire would behave there today: how fast it could spread, which direction it would head, and how intense it would be. It draws that spread right on the map so you can see it, not just read a number.

The app fills in everything automatically. It already knows what's growing at that spot, how steep the ground is, how dry things have been, and what the wind is doing, because it pulls all of that from free Canadian government data. You don't need to know anything about fire science to use it. Tools like this already exist in the US for firefighters. Nothing like it exists in Canada for regular people.

## How We Got Here

Our first idea was Smoke Signal, an app that warned people with asthma or COPD about wildfire smoke before it reached them. Our professor liked it, but wanted something less prediction-based. He pointed us to two American tools and asked what a Canadian version would look like.

**WindNinja** is a free program from the US Forest Service. A normal weather forecast gives one wind speed for a whole area, but real wind bends, speeds up, and swirls around hills, ridges, and valleys. WindNinja takes the forecast wind and reshapes it to fit the actual terrain. It gets used on nearly every large wildfire in the US, around 7 million times a year.

**Wildfire Analyst Pocket** is a phone app made by a company called Technosylva. A firefighter picks a spot, the app fills in the conditions (what's burning, wind, slope, how dry it is), and within seconds it shows how a fire there would behave: how fast it spreads, how long the flames are, and the shape of the burned area, drawn on a 3D map.

Both are built for fire crews, and both run on US data. That second part is the whole reason a Canadian version makes sense.

A note worth having ready for the professor: this new app still estimates something, but it's a different kind of estimate. Smoke Signal tried to guess what would happen in the future. This one is a calculator: "given today's real conditions, if a fire started here, this is how it would behave." It uses equations Canadian fire scientists built and tested on real fires, not a guess about tomorrow.

## Why a Canadian Version

Wildfire Analyst Pocket fills in what's growing at a spot using a US map called LANDFIRE, which stops at the border. It also runs American fire equations built around American vegetation. Canada has its own system: the Canadian Forest Fire Danger Rating System, which fire agencies here actually use. It has two parts:

- The **Fire Weather Index**, which rates how dry and fire-ready conditions are based on temperature, humidity, wind, and recent rain.
- The **Fire Behaviour Prediction system**, a set of equations that work out how fast a fire spreads and how intense it gets, for 16 Canadian vegetation types (black spruce, jack pine, aspen, open grass, and so on).

The government already publishes the ingredients for free: a national map of those vegetation types at 30m detail, daily fire weather maps, and live maps of active fires. So the science and the data exist. What's missing is an app that puts them together.

We checked what's already out there:

| Tool                    | Made by                  | What it does                                                                                            | What it doesn't do                                                          |
|-------------------------|--------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Wildfire Analyst Pocket | Technosylva (US)         | Map-based fire behaviour with auto-filled conditions                                                    | Canadian vegetation or Canadian equations; built for fire crews             |
| WindNinja               | US Forest Service        | Wind adjusted for terrain                                                                               | Fire behaviour itself; it's a wind tool for experts                         |
| FBP Go                  | Government of BC         | Canadian fire equations on a phone                                                                      | No map, no live weather, every input typed in by hand; built for fire crews |
| REDapp                  | Canadian fire agencies   | Canadian fire behaviour calculator                                                                      | Same as FBP Go: a calculator for trained users, not a map for the public    |
| CWFIS website           | Natural Resources Canada | National fire weather and active fire maps                                                              | Doesn't answer "what would a fire do at my spot"                            |
| <b>Our app</b>          | Us                       | Tap a spot, conditions fill in automatically, fire spread drawn on the map, explained in plain language | Doesn't replace official evacuation orders or professional fire tools       |

FBP Go matters most here, because it proves the Canadian equations can run on a phone. It just asks the user to already be a fire professional who knows the vegetation code for where they're standing. Our gap is the part in between: the map, the automatic data, and the plain-language result for someone who's never heard of a fuel type.

## Who It's For

People who aren't fire professionals but live with wildfire risk anyway:

- Cabin and cottage owners near forests.
- Farmers and ranchers, especially during spring grass fire season before things green up.
- Small towns, rural municipalities, and volunteer fire departments that don't have a fire behaviour specialist on staff.
- Campers, hikers, and anyone deciding whether today is a safe day for a campfire or burning a brush pile.
- Travellers heading somewhere fire-prone, like a summer trip to the Rockies (see What-If Mode and Trip Planning).

## An example

*The numbers below are made up to show how it would feel to use. They aren't real calculations.*

It's a dry Saturday in July. Dana owns a cabin north of Prince Albert and wants to burn a pile of brush she cleared last weekend.

She opens the app and taps her cabin on the map. Within a couple of seconds, the app has looked up everything about that spot on its own: the land around her cabin is mostly jack pine, it's on a gentle slope facing southeast, there's been no rain in 9 days, and the wind is coming from the northwest at 20 km/h.

The screen shows a coloured shape spreading out from her cabin, stretched toward the southeast, with three rings for 1 hour, 2 hours, and 4 hours. Underneath, in plain words: "Fire danger here today is High. A fire starting here could travel about 900 metres in the first hour, mostly toward the southeast, and would likely climb into the treetops."

A small icon also shows an active wildfire 40 km to the west, detected by satellite this morning, and the air quality layer shows smoke levels are still normal at her cabin.

Dana decides the brush pile can wait. She saves her cabin as a place to watch, so the app will tell her when fire danger there drops back to Low.

## How It Works, Step by Step

The app has two halves. One runs quietly in the background every day whether anyone opens the app or not. The other runs the moment someone taps a spot on the map.

Behind the scenes, every day

Collect data from  
government sources

Clean it and  
line it up

Store our  
own copy

When someone taps a spot

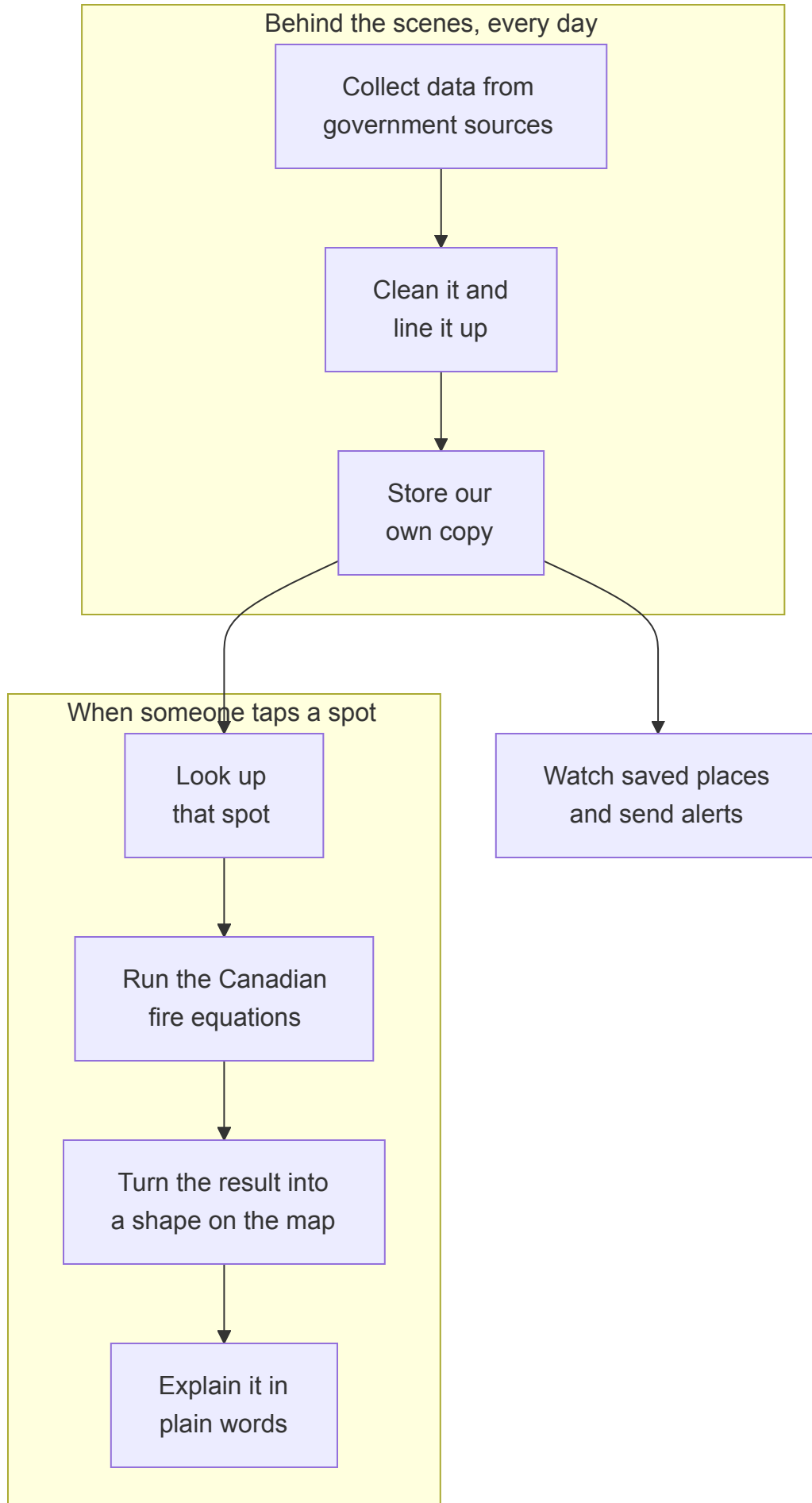
Look up  
that spot

Run the Canadian  
fire equations

Turn the result into  
a shape on the map

Explain it in  
plain words

Watch saved places  
and send alerts



## Behind the scenes

**1. Collect the data.** A scheduled job visits each data source on its own timetable and downloads what's new. The vegetation map barely changes, so we download it once a year. Fire weather updates daily. Wind and temperature update hourly. Satellite fire detections update several times a day during fire season. Nobody has to press a button; it just runs.

**2. Clean it and line it up.** Raw government data is messy, and this is the step that makes or breaks the app. A few real examples of what we have to fix:

- A weather station goes offline and reports nothing, so we fill the gap using the nearest working station and mark that it was filled in.
- One source gives wind in metres per second and another in km/h, so everything gets converted to the same units.
- The vegetation map stores codes like "C-3" that mean nothing to a normal person, so we translate them into names like "jack pine forest."
- Satellites sometimes flag things that aren't wildfires, like gas flares or factories that always run hot, so we filter out detections at known industrial sites.
- Every source uses its own map grid, so we line them all up so that one tap on the map pulls the right value from each source for the exact same spot.

**3. Store our own copy.** We keep everything in our own database instead of asking the government sites every time someone opens the app. That keeps the app fast, keeps it working if a government site goes down for an hour, and gives us history we can look back on.

## When someone taps a spot

**4. Look up that spot.** The app pulls four things for the exact location from our stored data: what's growing there, how steep the ground is and which way it faces, today's fire weather, and the current wind speed and direction.

**5. Run the Canadian fire equations.** Those four things go into the Fire Behaviour Prediction equations Canadian fire scientists built. Out come three answers: how fast a fire would spread, how intense it would burn, and whether it would stay on the ground or climb into the treetops (a crown fire, which is much more dangerous).

**6. Turn the result into a shape on the map.** Fires don't spread in circles. They stretch out in the direction the wind is pushing, like an egg with the pointy end facing downwind, and they run faster uphill. The app draws that stretched shape for 1, 2, and 4 hours so you can see how far and which way it would go.

**7. Explain it in plain words.** Numbers like "head fire intensity 4,000 kW/m" mean nothing to most people, so the app turns them into a danger level (Low, Moderate, High, Very High,

Extreme) and one or two plain sentences about what it means.

## For saved places

**8. Watch saved places and send alerts.** If someone saves their cabin or farm, the background job checks it every time new data comes in and sends a notification when the danger level changes or a new wildfire is detected nearby.

## What-If Mode and Trip Planning

This is the second feature, built once the core map works. The professor suggested it.

The core app answers "what would a fire do here today?" using real conditions. What-if mode lets you change those conditions yourself. Drag a slider to make the wind stronger, swing its direction, make it hotter or drier, and the spread shape on the map grows or shifts as you move it. Wildfire Analyst Pocket has this exact feature for fire crews. Ours would do it for regular people.

Trip planning ties the two together. You tell the app where you're going and when, say Calgary and then Banff:

- If the trip is in the next few days, the app fills in the forecast for each stop and flags any active fires near your route.
- If it's weeks away, nobody knows the weather yet, so the app opens what-if mode instead: here's what a fire near Banff would do on a typical dry, windy August day, and here are the sliders to see how bad it could get.

## An example

*Made-up numbers, for illustration only.*

Arjun is driving from Regina to Banff next week. He adds Calgary and Banff to a trip. The app shows fire danger is Moderate around Calgary and High near Banff, and marks an active fire about 60 km west of Banff that a satellite picked up yesterday.

He opens what-if mode for Banff and drags the wind from 15 to 40 km/h. The spread shape stretches to nearly three times its length and switches from a ground fire to a treetop fire. Now he understands why parks close trails on windy days, and he checks Parks Canada's fire updates before leaving.

## Why it's a good second feature

What-if mode runs the same Canadian fire equations as the core app, just with the person's numbers instead of the automatic ones. Once the core app exists, most of the hard work is

already done. It's also the clearest answer to what the professor asked for: a calculator for exploring "what if," not a prediction.

## Two things to get right

The screen has to make it obvious when you're looking at a made-up scenario rather than today's real conditions, something like a clear "What-if scenario, not a forecast" label, so nobody screenshots a slider experiment and treats it as real.

Mountains are also where our simple version is weakest. Banff sits in valleys that bend and funnel wind, which is exactly the problem WindNinja was built to solve. Our first version takes the forecast wind as-is, so shapes in the mountains will be rougher than on the prairies. If trip planning reaches the Rockies, the WindNinja-style terrain wind in the "Later" list becomes a lot more important.

## Fire History

Tap any spot, or open a saved place, and see where wildfires have actually burned around it over the past 50 years. This works the same in January as in July, so it gives people a reason to open the app all year.

**What someone sees.** Rings appear at 10, 25, and 50 km around the spot, with past fires shaded inside them, darker for more recent ones. A time slider filters by period, like "since 2000." Underneath, a plain summary (made-up numbers): "12 wildfires have burned within 25 km of your cabin since 1972. The closest was 8 km away in 2019. The largest burned about 40,000 hectares in 2015."

**Where the data comes from.** The National Burned Area Composite, Canada's official record of where wildfires burned, going back to 1972. It lives in the same CWFIS data catalogue we already use, so it's one new source, not a new system.

### How it works, step by step:

1. Collect: download every past fire outline once, then add each new season when the government publishes it each year.
2. Clean: merge fires recorded twice by different agencies, drop broken or impossibly tiny outlines, put every date in the same format, and mark fires before 1986 as less precise, since older satellites mapped them more roughly.
3. Store: save the outlines so the app can instantly answer "which fires touched this circle?" without searching 50 years of records on every tap.
4. On tap: draw the rings, find every fire that overlaps them, and work out distances, sizes, and years.
5. Explain: turn that into the short summary.

**Who it helps.** Someone buying or renting a cabin, a family choosing a campground for next summer, a town deciding where to focus clearing work.

**Being honest about it.** The records are strongest for large fires, so a small grass fire may not show up. And "no fires here in 50 years" doesn't mean a spot is safe, so the app should never word it that way.

**What's already out there.** CWFIS has its own fire history explorer. We couldn't see its features because the page loads dynamically, so someone should try it hands-on. Our difference is putting the history at your saved spot, right next to today's fire danger, instead of on a separate national map you'd need to know exists.

## Where the Data Comes From

Everything is free and public. A couple of sources need a free sign-up key, the rest don't.

| What we need to know                                   | Where it comes from                                                                                           | How often it changes                   |
|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|----------------------------------------|
| What's growing at each spot (vegetation type)          | <a href="#">National fuel type map</a> , Natural Resources Canada                                             | About once a year                      |
| How dry and fire-ready conditions are                  | <a href="#">Canadian Wildland Fire Information System</a> , Natural Resources Canada                          | Daily, all year                        |
| Wind, temperature, humidity                            | Environment and Climate Change Canada, with OpenWeatherMap as a backup                                        | Hourly                                 |
| How steep the ground is and which way it faces         | National elevation data, Natural Resources Canada                                                             | Practically never                      |
| Where wildfires are burning right now                  | Canadian Wildland Fire Information System (fire season, roughly May to September) and NASA's FIRMS satellites | Several times a day in season          |
| Smoke and air quality (add-on)                         | Environment Canada's Air Quality Health Index                                                                 | Hourly                                 |
| Where wildfires have burned in the past (fire history) | National Burned Area Composite, in the CWFIS data catalogue, Natural Resources Canada                         | Once a year, when each season is added |

Some of this carries straight over from the Smoke Signal research. NASA FIRMS, Environment Canada's air quality feed, and OpenWeatherMap were already checked out and confirmed to

work, so we're not starting from zero.

## First Version vs Later

| In the first version (by Week 12)                                                   | Later (ENSE 477 or beyond)                                                                                                |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Map focused on Saskatchewan, tap any spot                                           | All of Canada                                                                                                             |
| Vegetation, slope, fire weather, and wind filled in automatically                   | Wind adjusted for hills and valleys, the WindNinja side (it's free and open source, so it could plug in later)            |
| Canadian fire equations: spread speed, intensity, ground vs treetop fire            | Fire growth that changes as it crosses different vegetation and terrain, closer to Canada's professional Prometheus model |
| Spread shape on the map for 1, 2, and 4 hours                                       | Offline mode for cabins with no cell signal                                                                               |
| Danger level and a plain-language explanation                                       | French language support                                                                                                   |
| Active wildfires shown on the map                                                   | Health-specific smoke alerts, the original Smoke Signal idea                                                              |
| Accounts, saved places, and alerts when danger changes or a fire is detected nearby |                                                                                                                           |
| Smoke add-on: an air quality layer on the map                                       |                                                                                                                           |
|                                                                                     | What-if mode and trip planning, the first thing built after the core map                                                  |

One honest simplification in the first version: the spread shape assumes the vegetation and wind stay the same across the whole area. Real fires hit a lake, a road, or a different kind of forest and change behaviour. That's fine for a first version as long as the app says so clearly, and it's exactly what the "Later" column improves.

The smoke add-on stays deliberately small: a toggle that shows current air quality on the same map, using the Environment Canada feed we already researched. It helps the app stay useful outside fire season, since winter air quality in Saskatchewan has its own problems.

## Testing Outside Fire Season

We're building this in fall and winter, when almost nothing is burning. That's less of a problem than it sounds.

**Check our math against the official tools.** FBP Go and REDapp already run the same Canadian equations. Put the same inputs into their calculator and ours, and the answers should match. If they don't, we have a bug. This works any day of the year.

**Check our shapes against real past fires.** Canada keeps records of where past wildfires actually burned. We can pick a real fire from 2023, 2024, or 2025, feed the app the weather from those days, and compare the shape we draw with where the fire really went. It won't match perfectly, since our first version uses a simplified shape, but it should point the right way and be roughly the right size. That's also a strong thing to show evaluators.

**Test the daily pipeline live, right now.** Fire weather maps are published all year, not just in summer, so collecting, cleaning, and storing data can run for real from day one.

**Demo day.** Bazaar day is in late November, so the demo replays a real past fire day instead of relying on whatever is happening live.

## Things to Be Careful About

**This can't be mistaken for an evacuation tool.** Showing the public how a fire could spread is useful, but if someone treats our shape as "the fire will stop here," that's dangerous. The app has to say clearly that it doesn't replace official alerts and point people to the real ones (in Saskatchewan, the Saskatchewan Public Safety Agency and SaskAlert). Even FBP Go, built by the BC government for professionals, describes itself as a supplement to experienced judgment, not a replacement.

**The vegetation map can be wrong at a specific spot.** It's accurate for large areas, but a cabin with a cleared yard might still show up as "jack pine forest." Worth deciding whether users can correct the vegetation at their own spot.

**We're not the first Canadian fire calculator.** FBP Go and REDapp exist. Our answer, if asked, is the map, the automatic data, and the plain-language result for people who aren't fire professionals. Better to say that upfront than get caught out.

**We shouldn't invent our own danger levels.** CWFIS already publishes official fire danger classes (Low through Extreme). Same principle as Smoke Signal: reuse the official wording rather than making up our own.

## Open Questions for the Team

- What do we call it?
- Web app, phone app, or both for the first version?

- Which exact Environment Canada feed do we use for wind and humidity? Needs a real test call.
- Do we let users correct the vegetation type at their own spot?
- Who do we interview for user research? Cabin owners, a rural municipality, a volunteer fire department?
- Can we get 20 minutes with someone who does fire behaviour work professionally, to sanity-check our results?
- Trip planning to Calgary and Banff means going past Saskatchewan. All our data sources are national, so do we cover the three Prairie provinces from the start instead of Saskatchewan only?
- Fire history mostly reuses what we already collect. Does it go in the first version, or right after alongside what-if mode?

## Sources

- [WindNinja](#), US Forest Service Missoula Fire Sciences Lab
- [Wildfire Analyst Pocket Edition](#), Technosylva
- [FBP Go](#), Government of British Columbia
- [REDapp demonstration](#), Fire Research and Management Exchange System
- [Canadian Forest Fire Behavior Prediction System](#), Canadian Wildland Fire Information System
- [Canada's Fire Behaviour Prediction System fuel types](#), Natural Resources Canada
- [FBP Fuel Types 2024, 30m](#), Open Government Portal
- [CWFIS Datamart](#), Natural Resources Canada
- [Development and structure of Prometheus](#), the Canadian wildland fire growth model
- [National Burned Area Composite downloads](#), CWFIS
- [CWFIS fire history explorer](#), Natural Resources Canada